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Bachelor of Computer Application

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Title of the project: E-VOTING

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ABOUT

Electronic voting (e-voting) revolutionizes traditional ballot-casting processes by employing digital platforms such as internet-based systems, electronic voting machines (EVMs), and mobile applications. Proponents argue that e-voting enhances accessibility, efficiency, and accuracy in elections, potentially boosting voter turnout and expediting result tabulation. However, concerns persist regarding cyber security vulnerabilities, ensuring equitable access to technology, and maintaining the integrity of the democratic process.

SYNOPSIS

Abstract:

Electronic voting (e-voting) systems have emerged as a promising solution to modernize the electoral process, leveraging digital technology to streamline voting procedures, to help physically challenge people, to avoid scam and potentially increase voter participation.

Introduction:

- The main agenda of e-voting is to avoid scams through one time login system.
- To help physical challenge people in a way that they can vote from home through mobile phones.

Key features of e-voting include:

Accessibility: E-voting systems aim to make the voting process more accessible to a wider range of voters, including those with disabilities, individuals living in remote areas, and citizens residing abroad. By offering electronic platforms, voters can cast their ballots from any location with internet access or through electronic voting machines deployed in polling stations.

Efficiency: E-voting streamlines the voting process, reducing the time and resources required for ballot distribution, collection, and counting. Electronic systems facilitate quick and accurate tabulation of votes, potentially leading to faster declaration of election results.

Accuracy: Electronic voting systems minimize the risks associated with human error in manual vote counting. By automating the tallying process, e-voting helps ensure greater accuracy and reduce the likelihood of discrepancies or miscounts in election results.

Security: Security measures are crucial components of e-voting systems to safeguard against cyber threats, hacking, tampering, and fraud. Strong encryption protocols, authentication mechanisms, and audit trails are implemented to protect the integrity and confidentiality of the voting process and maintain trust in the electoral system.

Transparency: E-voting systems promote transparency by providing visibility into the entire voting process, from ballot casting to result tabulation. Through features such as digital receipts, verification codes, and public audit trails, voters can verify that their votes are recorded accurately and counted correctly, enhancing trust and confidence in the electoral process.

Flexibility: Electronic voting offers flexibility in terms of voting methods and timing. Voters can choose to cast their ballots remotely through online platforms or in-person using electronic voting machines, and some e-voting systems allow for extended voting periods, accommodating diverse schedules and preferences.

Auditability: E-voting systems incorporate auditability features that enable independent verification of election results. Through post-election audits, election authorities and stakeholders can review and validate the integrity of the voting process, identifying any irregularities or discrepancies for investigation and resolution.

EXISTING SYSTEM

Internet-based e-voting systems: These systems allow voters to cast their ballots remotely using internet-connected devices such as computers, tablets, or smartphones. Voters access a secure online platform where they authenticate their identity and submit their votes electronically. Countries like Estonia, Switzerland, and some regions in the United States have experimented with internet voting systems for certain elections.

Electronic voting machines (EVMs): EVMs are standalone devices deployed at polling stations to facilitate electronic voting. Voters use the machines to make their selections electronically, typically through a touchscreen interface or by pressing physical buttons. Once voting is complete, the EVMs store and tally the votes digitally. Various countries, including India, Brazil, and the Philippines, use EVMs extensively in their elections.

Hybrid e-voting systems: Some e-voting systems combine elements of internet-based voting with traditional paper-based methods. For example, voters may have the option to cast their ballots electronically or by using a paper ballot at designated polling stations. These hybrid systems aim to offer flexibility while also addressing concerns about cyber security and voter trust.

PROPOSED SYSTEM

Authentication: Implement a robust authentication process to verify the identity of voters. This may include biometric authentication (such as fingerprint or facial recognition)

User and Admin login: Implementing multiple login.

FUTURE ENHANCEMENT

Biometric authentication: Implement advanced biometric authentication methods such as iris scanning or voice recognition to further enhance the security and accuracy of voter verification, reducing the risk of identity fraud.

Holomorphic encryption: Utilize holomorphic encryption techniques to allow for the secure computation of election results without decrypting individual votes, maintaining the anonymity of voters while ensuring the integrity of the tallying process.

Mobile voting innovation: Develop innovative mobile voting solutions that leverage emerging technologies such as 5G networks, edge computing, and secure enclave technologies to enable seamless and secure voting experiences on smartphones and other mobile devices.

CONCLUSION ON PROPOSED SYSTEM

In conclusion, the incorporation of robust authentication mechanisms and multiple login functionalities is imperative for ensuring the integrity and security of e-voting systems. By leveraging biometric authentication, such as fingerprint or facial recognition, the identity of voters can be reliably verified, mitigating the risks of fraudulent activities and enhancing the overall trustworthiness of the electoral process.

Furthermore, the implementation of separate user and admin logins adds an additional layer of security, facilitating controlled access to sensitive administrative functions and preventing unauthorized access. Together, these measures contribute to the establishment of a secure and transparent e-voting environment, fostering confidence among voters and election stakeholders in the fairness and reliability of the electoral process.

LITERATURE SURVEY

A literature survey on e-voting reveals a diverse range of research encompassing various aspects of electronic voting systems, including technological innovations, security concerns, usability issues, legal frameworks, and social implications. Here is a brief overview of the key findings and themes identified in the literature

Scholars analyze the legal and regulatory frameworks governing e-voting in different countries and jurisdictions, examining issues such as voter privacy, data protection, election integrity, and the role of electoral authorities in overseeing e-voting processes

Many literature reviews include case studies and evaluations of e-voting implementations in real-world elections, assessing the success, challenges, and lessons learned from different deployment scenarios. Researchers analyze factors contributing to the effectiveness and reliability of e-voting systems, as well as instances of failure or controversy.

Software Requirements Specification (SRS)

OPERATING SYSTEM	Popular operating systems like Windows 10 pro, mac OS, or Linux distributions are all suitable.
PHP (version 8.1.3)	PHP (Hypertext Preprocessor) is a widely used open-source server-side scripting language primarily designed for web development but also used as a general purpose programming language.
My SQL	MySQL is an open-source relational database management system (RDBMS) that is widely used for managing structured data. It is developed, distributed, and supported by Oracle Corporation. MySQL is known for its reliability, scalability, and performance, making it a popular choice for many web applications and services
XAMPP (version 8.2.12)	XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, Maria DB database, and interpreters for scripts written in the PHP and Perl programming languages.

Hardware Requirement Specification

Processing Power	A CPU with 8 to 16 cores or more would be suitable for such a system, depending on the workload and performance requirements. With respect to demo of project Intel(R) Core(TM) i5-2450M CPU @ 2.50GHz 2.50 GHz
Storage	Based on this hypothetical scenario, the estimated RAM requirement for the e-voting system would be approximately 708 GB. With respect to demo of project 4GB RAM is needed.

Functional Requirement

Voter Registration: The system should allow eligible voters to register securely, providing necessary personal information and verifying voter eligibility according to electoral rules and regulations.

Vote Casting: The system should facilitate the secure casting of votes by registered voters, allowing them to make selections on electronic ballots using intuitive interfaces, and ensuring vote confidentiality and integrity.

Vote Tallying: The system should accurately tally and aggregate votes cast by voters, providing real-time or near-real-time updates on election results and ensuring the integrity and accuracy of the tallying process.

Auditability: The system should maintain comprehensive audit trails and logs to enable post-election audits, ensuring transparency, accountability, and verifiability of election results.

Security: The system should implement robust security measures to protect against unauthorized access, tampering, and fraud, including encryption, authentication, access control, and intrusion detection mechanisms.

Accessibility: The system should be accessible to voters of all abilities and disabilities, providing alternative voting methods and interfaces to accommodate diverse user needs and ensure equal access to the electoral process.

Non-Functional Requirements:

Reliability: The system should be reliable and available throughout the duration of the election period, minimizing downtime, disruptions, and system failures to ensure continuous access to voting services.

Performance: The system should exhibit optimal performance, handling peak loads of concurrent users and processing votes efficiently to maintain responsiveness and scalability under heavy voting traffic.

Scalability: The system should be scalable to accommodate fluctuations in voter turnout and handle increasing data volumes without compromising performance, reliability, or security.

Usability

Compatibility

Data Privacy

Database Requirements:

Tables: -

Voter: (voter name, voter ID)

Admin: (admin ID, admin password)

Party: (party name, candidate name, candidate ID)

Interfaces: -

Web Interface: -

HTML/CSS/JavaScript for the frontend.

PHP for server-side scripting.

Database Interface: -

MySQL for database management.

SQL queries for data manipulation
